Week 10 Extra In-Class Exercises

(more on files)

Q1: Grades

You are given a file called "grades.txt" that contains records of students' grades of different courses. Each line of the file contains a student's name, the course he/she took, the term he/she took the course and the grade he/she received for that course. The four columns are separated by the character `'#'`. Open the file to understand the structure of the file.

Part (a) [**/***]

Write a program that creates a new file called `'course_info.txt'` that contains the names of students who received As for each course. If a course doesn't have any student who received A, that course should not be included in the file. The generated file should look something like the following:

```
IS101
Wendy Li
Joe Wong
Ng Yin Hui
Chua Lee Hong

IS111
Wendy Li
Ng Yin Hui
Peter Liu

IS112
Joe Wong
Peter Liu
Eric Wong
```

Part (b) [***]

Write a program that prints out all pairs of students who have taken the same course together (i.e., during the same term) at least once. If two students have taken more than one course together, they should be displayed only once.

With the file "grades.txt" we provide, your program should display the following information:

```
Wendy Li, Joe Wong
Wendy Li, Peter Liu
Joe Wong, Peter Liu
Joe Wong, Eric Wong
Ng Yin Hui, Chua Lee Hong
Ng Yin Hui, Eric Wong
```

Note that the pairs may appear in a different order and also the pair (A, B) is the same as the pair (B, A).

Q2: Most Frequent Words [***]

You are given a file that contains a news article. Write a program that reads this file and finds out which **three** words appear most frequently in the article. The program should print out the top-3 most frequent words and their corresponding counts. (You can assume that the article contains at least three different words.)

You should read the file line by line and split each line into individual words by spaces. You should change everything to lowercase before counting the words. You do not need to remove punctuation marks. E.g., if a sentence is “That’s a great idea.” we consider it to have 4 words: “that’s”, “a”, “great”, “idea.” (Note that the last word contains the full stop.)

We’ve provided you with two files: 'article-1.txt' and 'article-2.txt'.

- For 'article-1.txt', the most frequent three words are

```
the: 46
to: 29
singpass: 18
```

- For 'article-2.txt', the most frequent three words are

```
and: 20
the: 16
to: 14
```

Hint: You can use a list `[w, c]` to store the count `c` of the word `w`, where `c` can be updated while you’re reading the file. You can store the counts of all words as a list of lists.

(Note: Next week you’ll learn about dictionaries, which will allow you to solve this same problem with a better solution.)